

Continuing Professional Development Log

Name: Tyler Ward

Current and recent CPD activity:

CPD Activity Title	Description	Dates	CPD Hours
OpenGL and Real-Time Rendering	Self-taught OpenGL, shaders, and 3D rendering concepts using online resources.	Aug 2025 – Oct 2025	~20 hours
Haptic Rendering Methods	Read academic papers on virtual coupling and proxy-based haptics to understand force rendering approaches.	Aug 2025 – Jan 2026	~20hrs
BLDC Motor Control (SimpleFOC)	Learned to use SimpleFOC library for BLDC motor control and torque-mode operation.	Jan 2026 – Mar 2026	~10 hours
PhysX Simulation Integration	Learned how to integrate and use a physics engine (PhysX) for real-time simulation.	Jan 2026 – Feb 2026	~10hrs
Hardware Integration and Iterative Design	Gained practical experience designing and refining a mechanical system with motors and encoders.	Nov 2025 – Mar 2026	~20hrs
Working with a Large Codebase	Gained experience developing and managing a multi-component C++ codebase, including structuring modules, debugging across subsystems, and maintaining clarity in a multi-threaded architecture.	Aug 2025 – Apr 2026	~50 hrs
Technical Report Writing and Literature Review	Developed skills in structuring a technical report and conducting a focused literature review, including synthesising multiple sources, strengthening technical claims with references, and improving clarity and precision in written communication.	Dec 2025 – May 2026	~25 hrs

Planned and Future CPD activity:

CPD Activity Title	Description	Skills addressed	Dates
Technical Research and Literature Skills	Further develop ability to read and interpret technical research papers, including understanding methodologies, extracting key insights, and critically evaluating approaches.	Research skills, critical analysis, technical understanding	Summer 2026
Long-Term Project Planning	Improve ability to plan and manage longer-term technical projects, including setting realistic milestones, managing uncertainty, and adapting plans as challenges arise.	Project management, planning, problem-solving	Summer 2026